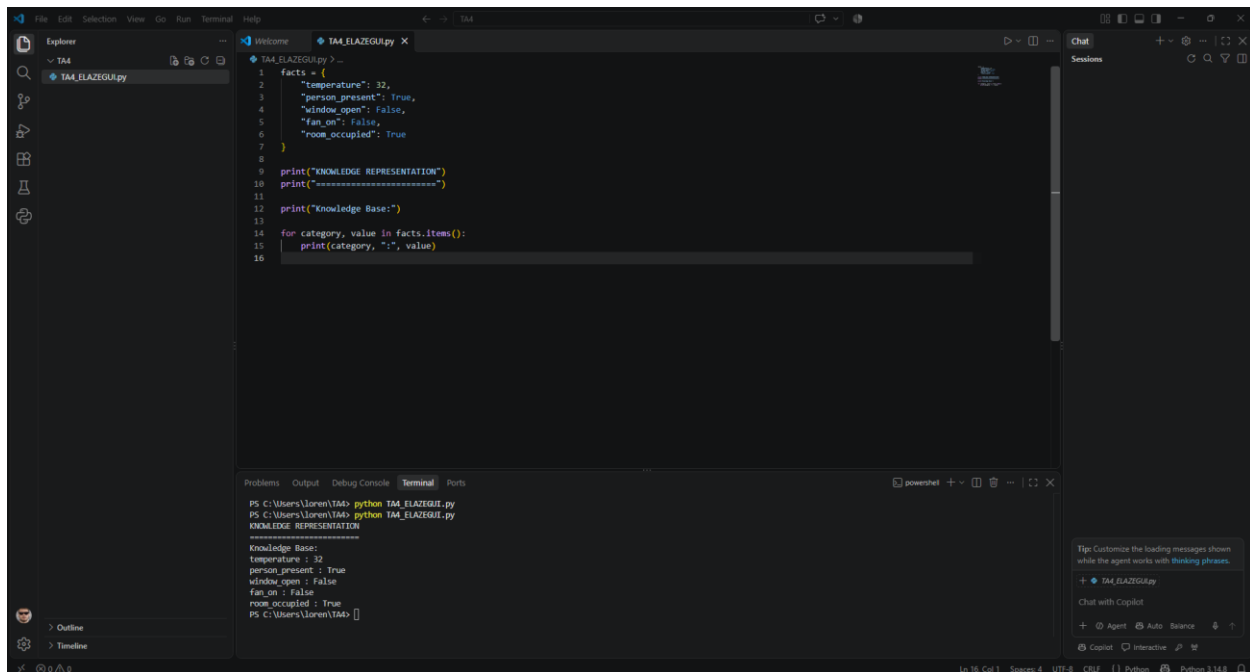


Knowledge Base and Use Case:

The knowledge base contains information about a room, including its temperature, whether a person is present, whether the window is open, the current fan status, and whether the room is occupied. The system uses these facts to determine appropriate actions for controlling the fan. The use case demonstrates how Knowledge Representation, Rule-Based Reasoning, and Case-Based Reasoning can be applied to a simple smart-room environment.

1. Knowledge Representation / Knowledge Base



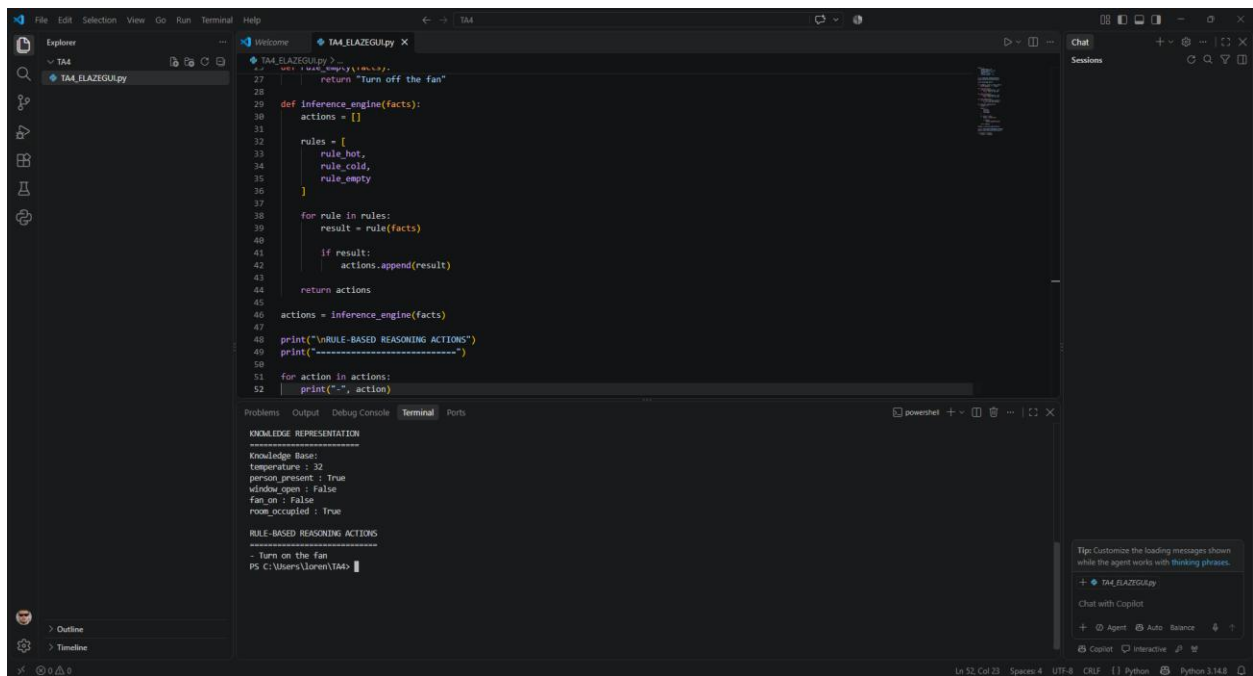
The screenshot shows a Visual Studio Code editor with a Python file named `TAM_ELAZEGUI.py`. The script defines a dictionary of facts and prints them out. The terminal output shows the execution of the script, displaying the facts and the knowledge base representation.

```
1 facts = {  
2     "temperature": 32,  
3     "person_present": True,  
4     "window_open": False,  
5     "fan_on": False,  
6     "room_occupied": True  
7 }  
8  
9 print("KNOWLEDGE REPRESENTATION")  
10 print("=====")  
11 print("Knowledge Base:")  
12  
13  
14 for category, value in facts.items():  
15     print(category, ":", value)  
16
```

Terminal Output:

```
PS C:\Users\loren\TAM> python TAM_ELAZEGUI.py  
PS C:\Users\loren\TAM> python TAM_ELAZEGUI.py  
KNOWLEDGE REPRESENTATION  
=====  
Knowledge Base:  
temperature : 32  
person_present : True  
window_open : False  
fan_on : False  
room_occupied : True  
PS C:\Users\loren\TAM>
```

2. Rule-Based Reasoning (RBR) Output



```
27         return "Turn off the fan"
28
29     def inference_engine(facts):
30         actions = []
31
32         rules = [
33             rule_hot,
34             rule_cold,
35             rule_empty
36         ]
37
38         for rule in rules:
39             result = rule(facts)
40
41             if result:
42                 actions.append(result)
43
44         return actions
45
46 actions = inference_engine(facts)
47
48 print("\nRULE-BASED REASONING ACTIONS")
49 print("-----")
50
51 for action in actions:
52     print("-", action)
```

KNOWLEDGE REPRESENTATION

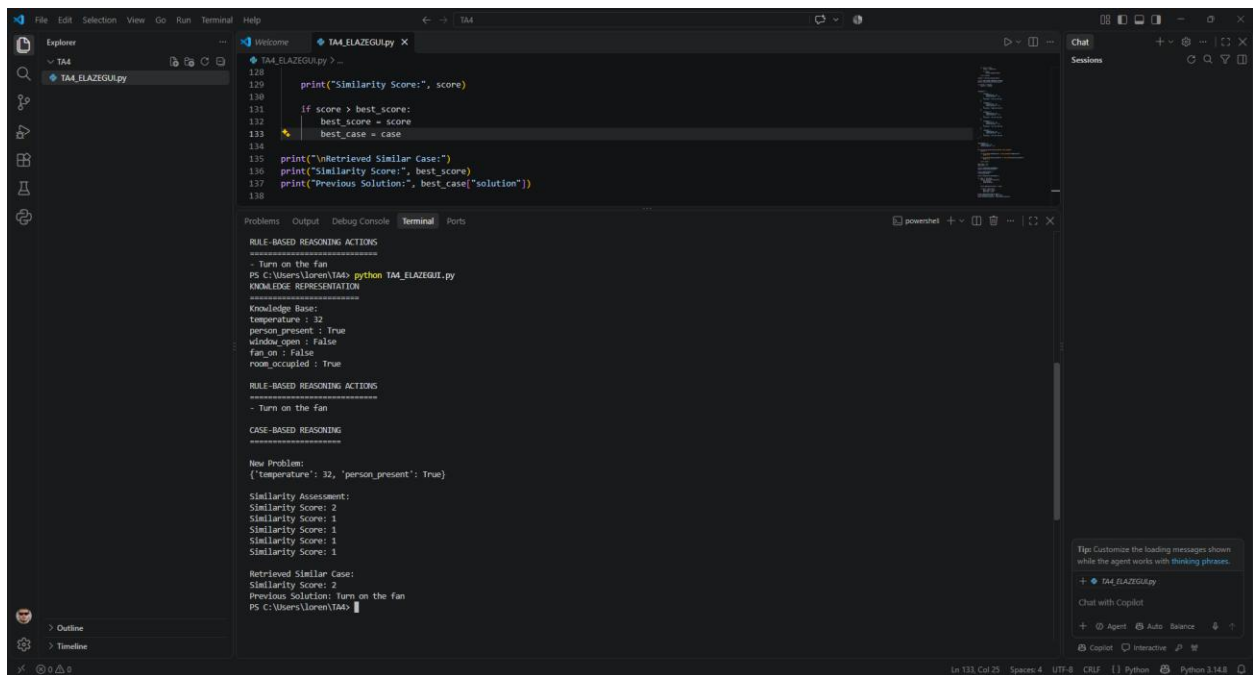
Knowledge Base:

temperature : 32
person_present : True
window_open : False
fan_on : False
room_occupied : True

RULE-BASED REASONING ACTIONS

- Turn on the fan
PS C:\Users\loren\IAD>

3. CBR Similarity Assessment + Retrieved Case



```
128     print("Similarity Score:", score)
129
130     if score > best_score:
131         best_score = score
132         best_case = case
133
134     print("\nRetrieved Similar Case")
135     print("Similarity Score:", best_score)
136     print("Previous Solution:", best_case["solution"])
137
138
```

RULE-BASED REASONING ACTIONS

- Turn on the fan
PS C:\Users\loren\IAD> python TAI_ELAZEGUI.py

KNOWLEDGE REPRESENTATION

Knowledge Base:

temperature : 32
person_present : True
window_open : False
fan_on : False
room_occupied : True

RULE-BASED REASONING ACTIONS

- Turn on the fan

CASE-BASED REASONING

New Problem:

{'temperature': 32, 'person_present': True}

Similarity Assessment:

Similarity Score: 2
Similarity Score: 1
Similarity Score: 1
Similarity Score: 1
Similarity Score: 1

Retrieved Similar Case:

Similarity Score: 2
Previous Solution: Turn on the fan
PS C:\Users\loren\IAD>

4. CBR Reuse + Revise + Retain + Updated Case Base

